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Towards a fair revenue distribution of a Renewable Energy Community through a proportional energy consumption model application

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Abstract. In recent years, the frequency of extreme events such as floods, droughts, and heat waves has been increasing due to climate change. To mitigate and reduce the devastating effects of these phenomena, finding a common solution that directly involves citizens in the energy transition process is crucial. In this context, the European Union, through the Green Deal, has set the ambitious goal of achieving carbon neutrality by 2050, and, through the REPowerEU plan, covering at least 42.5% of total energy consumption with renewable energy sources by 2030. The renewable energy communities, introduced by the EU Directive 2018/2001, represent a key tool for achieving these European targets. Renewable energy communities are based on a bottom-up business model as they directly engage citizens and facilitate the expansion of renewable energy generation. Moreover, in several countries, including Italy, economic support schemes have been introduced to incentivize the development of these energy-sharing configurations. This paper analyzes a specific fair revenue distribution model among community members, taking the renewable energy community in Benevento (Southern Italy) province as a case study. The analysis highlights how variations in user types and energy consumption impact individual contributions. This underscores the need for an economic distribution model with a dynamic and flexible structure, allowing adaptation to different configurations. Therefore, the used algorithm should incorporate specific parameters that account for the different operating conditions, such as the presence of third-party producers, prosumers, and high-energy consumers, as well as solidarity-based objectives, such as addressing energy poverty. All these factors are crucial, as the implementation of an inaccurate economic distribution model could favor some users at the expense of others and, at the same time, promote environmentally and energetically unsustainable behaviors.

1. Introduction

As reported by the International Energy Agency (IEA), the growth rate of Carbon Dioxide (CO₂) emissions is decreasing due to the significant efforts made over the last decade to mitigate the causes of climate change [1]. However, this reduction is still insufficient to limit the increase in global mean temperature to below 1.5°C, compared to pre-industrial levels, as set by the Paris Agreement [2]. In fact, 2023 was marked by a higher number of days on which the temperature rises exceeded the 1.5°C threshold [3]. Climate change is not solely an environmental problem, its consequences affect all aspects of human life, undermining fundamental rights such as the right to a dignified existence. Therefore, climate change is also a social, economic, and political problem and requires a holistic approach to be addressed.

Due to its heavy dependence on fossil fuels, the energy sector is one of the key areas in which action is needed: currently, three-quarters of greenhouse gas emissions (GHG) come from the



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energy supply chain [4]. Since the growth of the population, especially in emerging and low-income countries [5], and the increasing energy consumption of the computing systems, due to Artificial Intelligence (AI), this ratio will continue to rise without the implementation of renewable energy-based plants. In addition, to be effective, the energy transition must involve citizens in a collective action: it is necessary a shift from a centralized energy supply chain to a distributed one, thanks to the figure of the prosumer, individuals who both produce and consume energy.

The Renewable Energy Community (REC) represents a valid tool to achieve all these goals: it is based on distributed Renewable Energy Sources (RES) plants, such as rooftop photovoltaic (PV) systems, small-scale wind turbines, etc., and encourages the active participation of citizens through a democratic, bottom-up administrative model [6]. Moreover, thanks to national legislative support schemes, RECs could help to address social and solidarity issues, such as Energy Poverty (EP) [7] and access to renewable energy, while helping to overcome the public resistance to adopting RES technologies.

RECs and the more general concept of energy community were introduced in the Clean Energy for All Europeans Package (CEP) 2019 through the Renewable Energy Directive 2018/2001 (RED II) and the Internal Electricity Market Directive 2019/944 (IEMD) [8]. Article 21 of RED II introduced the concept of collective self-consumption of renewable energy, which is the core of RECs. Indeed, REC is defined as a legal entity:

- *based on open and voluntary participation, is autonomous, and is effectively controlled by shareholders or members that are located in the proximity of the renewable energy projects that are owned and developed by that legal entity;*
- *the shareholders or members of which are natural persons, SMEs, or local authorities, including municipalities;*
- *the primary purpose of which is to provide environmental, economic, or social community benefits for its shareholders or members or for the local areas where it operates, rather than financial profits.*

Members of REC can:

- *produce, consume, store, and sell renewable energy, including through renewables power purchase agreements;*
- *share, within the renewable energy community, renewable energy that is produced by the production units owned by that renewable energy community [9].*

In addition, the European Union (EU) requires Member States to include support schemes in their energy policies to enhance the spread of RECs and, generally, energy generation from RES.

Several Countries, such as Italy and Germany, have developed a comprehensive legislative framework, including support schemes, that enable the establishment of RECs; others have only defined a partial regulatory framework [10]. However, the flexibility ensured by the EU to Member States in transposing EU Directives into national legislative frameworks has led to the emergence of several differences in the definition of RECs, particularly in terms of geographical boundaries, maximum capacity of generation units, both individually and collectively, and the design of incentive schemes. Consequently, the academic literature reflects a wide range of research topics, from technical optimisation to environmental assessment, including social aspects, such as EP or integration of industrial districts within the near urban area, grid stability, and ancillary services [11], [12]. Nevertheless, the fair distribution of incentives among members of RECs, based on the chosen business model, is rarely assessed in the literature.

For instance, in [7] a comprehensive assessment is conducted that includes both 3E (Energy, Economic, Environmental) analysis and energy poverty considerations for a REC composed of three residential users in Southern Italy. The authors show that the support schemes at the base of RECs could help reduce the payback period (SPB) of the initial investment cost of PV systems and overcome the EP issue for its members. For evaluating EP conditions, the study introduces a novel index based on ISEE (Indicatore della Situazione Economica Equivalente), an Italian economic index used by institutions to assess socio-economic eligibility for social benefits. Without RECs, according to a 10%_{ISEE} index, users #2 and #3 are in EP conditions since their energy expenditure is more than 10% of relative ISEE. In the REC configuration, the 10%_{ISEE} index of members #2 and #3 drops from 10.41 and 10.93% to 6.58 and 5.25%, respectively; therefore, no users are in EP condition. Despite these promising results, the study does not address models for the fair distribution of economic incentives within the REC.

1.1. Novelty

The aim of the paper is to demonstrate how an accurate assessment of economic sharing models can help to improve the EP conditions of members of RECs, starting from the incentive support scheme set by the Italian regulatory framework. Since RECs are based on a virtual model, as detailed below, that enables the energy sharing among RECs' end-users, only prosumers are effectively characterised by a physical self-consumption, and therefore, get discounts on their bills. Consumers, although the energy shared within RECs, still bill the same amount of energy; thus, they pay bills as usual.

In addition, the economic support scheme set by the Italian legislative framework also provides some financial incentives, such as tax deductions and capital grants. However, these are valuable tools for users who, at least partially, have the financial capability to cover part of the investment cost. On the other hand, users in the EP condition would still find it difficult to take advantage of the tools available. Therefore, the greatest advantages, relative to participation in a REC, are for prosumers, not consumers, and, in general, for users in an EP condition. Thus, the only way a REC can truly improve the conditions of people in EP is through a wise choice of the economic distribution model of the incentives earned within the community.

As shown, the fair distribution of RECs' revenues among members is a crucial aspect for RECs, but, despite its importance, is rarely addressed in the literature. Most studies are focused on cooperative game theory; however, this approach is feasible only for RECs with few members [13], [14]. The computation of the Shapley value requires solving an optimization problem whose computational time grows exponentially with the number of participants, since the algorithm applied explores all 2^N possible combinations (with N number of end-users) to ensure the fairness of the found solution [15].

In addition to scalability issues, the fair economic-sharing schemes must consider the absence of a specialised figure able to handle complex models for the computation of shared energy by each member and the relative incomes, especially in the case of RECs promoted by a small municipality, where administrative and technical resources are often limited. Implementing and managing RECs in real-world contexts requires a more practical approach than is usually assumed in the academic literature. In particular, the absence of specialized technical personnel, especially in RECs promoted by small municipalities or charitable organisations, is a critical barrier that is frequently underestimated.

This study aims to develop an economic incentive distribution model that addresses all the requirements. The proposed model is proportional to the energy consumption of the end-users

through a weighted average, but, at the same time, by means of multiple factors, it also addresses solidarity purposes.

To assess the model's validity, it was applied to a real REC in Campania (Southern Italy), promoted by the local municipality. The main goal of this case study is to demonstrate how the proposed economic distribution model, which also pursues solidarity-oriented goals, must dynamically adapt to varying compositions of REC members to prevent instability-related issues. In particular, the study seeks to illustrate how, by appropriately tuning the parameters of the proposed economic model, RECs can actively contribute to mitigating EP among their members. In this case, due to the unavailability of the ISEE data of end-users, members with unitary electricity costs billed (C_u) higher than the median value of members' REC are considered potential users in an EP condition, since their energy expenditures are significant.

The Gini Index, the average gain of users in the potential EP condition / non-EP condition, and the remuneration per unit of shared energy for each member are computed to validate and compare the results. These indicators offer a comprehensive evaluation framework to determine how the proposed model can ensure equitable and stable economic returns while supporting vulnerable users within the community.

1.2. Organization

The introduction section is followed by the materials and methods section. First, an overview of the incentives support scheme established by the Italian regulatory framework is reported, and then the energy characterization of REC load and PV production and the definition of operative scenarios are presented. Finally, the section ends with the mathematical formulation of the economic sharing model and evaluation indices used, such as the Gini index. In the third section, the results obtained are reported. The last section contains final considerations, ideas, and suggestions for future developments.

2. Materials and Methods

This section reports a schematic overview of the Italian regulatory framework to clarify the definition of shared energy and the relative computation of the incentive. Then, the mathematical formulation of the benefits-sharing model is reported.

2.1. Italian regulatory framework and economic support scheme

Despite the long time required by the Italian authorities to transpose the RED II into national law, Italy is among the few European countries that have developed a complete legislative framework for the establishment of RECs:

- The UE Directive 2018/2001 was transposed through Legislative Law 199/2021 [16];
- The virtual model, which defines how renewable energy is shared among REC members, is regulated by TIAD, published by ARERA (Autorità di Regolazione per l'Energia Reti e Ambiente) [17];
- The economic support scheme is regulated by the Ministerial decree 414/2023 [18].

According to the current regulations, RECs are defined as a group of end-users who share renewable energy produced by local distributed RES-based plants. However, there are some limitations:

- Each end-user is identified by a unique POD (Point of Delivery). Only members whose POD is connected to the same primary substation are eligible to join the same REC;
- The maximum nominal capacity allowed for each generation unit is less than 1 MW;

- The shared energy among members of the REC is computed with an hourly time step resolution, and it corresponds to the minimum value between the exceeded renewable energy produced by the local RES-based plant fed into the grid and the simultaneous energy withdrawals from the same low-voltage power grid.

Due to the implementation of the virtual model, virtually all community members can self-consume renewable energy produced by local plants, even if not everyone owns a RES plant. Share energy within REC is referred to as *collective self-consumption*, but it is different from on-site self-consumption since there is no direct connection between the withdrawal and production points. [19]. Therefore, all consumers, although the energy shared within RECs, still bill the same energy, thus they pay bills as usual. Only prosumers, thanks to physical self-consumption, get discounts on their bills.

On the other hand, regarding shared energy, Italian economic support schemes defined two types of tariffs:

- TIP (Tariffa Incentivante Premio): it is an incentive-based tariff composed of a fixed component that depends on the nominal capacity of the RES-based plant, and a variable component that depends on the electricity market price in the relevant bidding zone (Pz). Moreover, an incrementation of + 4 €/MWh and + 10 €/MWh is applied for PV systems located in Central and Northern Italy, respectively[18];
- Cost-reflective Tariff: ARERA has established a fixed-value tariff for shared energy within the community that reflects the mitigation of energy flow impact on the power grid, as the energy produced by the distributed RES-based plants is withdrawn in the same local portion of the grid [17], [20].

In addition, the GSE (Gestore dei Servizi Energetici), through the RID (Ritiro Dedicato) mechanism, gives a remuneration at Pz to users who feed into the grid the energy produced by the RES-based plants that is not used for on-site self-consumption [20].

2.2. Case Study

In the following section, an energy characterization of REC, called GREEN CER Campolattaro, which is used to validate the effectiveness of the adopted economic benefits-sharing model, is reported. GREEN CER Campolattaro is a public-led energy community in Campolattaro (430 m a.s.l, 1853 Heating Decree Days, and climatic zone D). The REC includes 30 members; however, the users provide only monthly energy consumption data. Consequently, their hourly load curves, needed for calculating shared energy, were reconstructed using a standardized hourly percentage profile, as described in detail in a later section. Due to the limited variability among the reconstructed profiles (which would have led to overlapping hourly curves), it was not meaningful to consider all members in the analysis. Therefore, only a subset of members was selected for the validation test of benefits-sharing models.

Table 1 reports the key information about the selected user's electricity consumption, as retrieved from their energy bills. According to the Italian regulation on electrical bills, end-users are categorized based on the type of electricity contract as *domestic resident* and *other usage*. In the first case, the titular of electrical furnishing has the official residence at the same address as the POD, while other usage is applied to non-domestic contracts that are usually associated with commercial or service-oriented activities.

Each POD is also associated with a nominal power value, representing the maximum power drawn from the grid under the contractual agreement. This data can help to understand

Table 1. Summary of electricity consumption data for selected users, categorized by market type, contracted power, and annual energy usage.

User number ID	User types	Contractual bill Power [kW]	Type of bill	PV system	Total annual consumption [kWh/year]	Cu [c€/kWh]
#1	Domestic Resident	6	Mercato libero	Yes	2595	49.5
#2	Domestic Resident	6	Mercato libero	No	3000	39.6
#3	Domestic Resident	3	Mercato libero	No	1376	25.8
#4	Domestic Resident	3	Mercato libero	No	2312	40.6
#5	Domestic Resident	3	Maggior tutela	No	3260	56.5
#6	Domestic Resident	3	Maggior tutela	No	2615	59.6
#7	Domestic Resident	6	Maggior tutela	No	2614	30.3
#8	Domestic Resident	3	Mercato libero	No	1476	44.6
#9	Domestic Resident	3	Mercato libero	No	3500	14.8
#10	Domestic Resident	3	Mercato libero	No	3339	39.4
#11	Domestic Resident	3	Mercato libero	No	2948	32.4
#12	Other usage (City Hall)	4.5	Mercato libero	Yes ^a	9048	40.3

^a It will be installed with 7 kWp.

the types of appliances used by the different members. For instance, in the case of induction cooking, the contractual nominal power is higher when this appliance is not present.

Finally, users may be served under either the *mercato libero* (liberalized market) or the *mercato di maggior tutela* (regulated market), depending on their contractual choices.

To test the flexibility of the benefits-sharing model, two scenarios are considered:

- *Scenario A:* the REC load is given by the energy demand of end-users with IDs from #1 to #11;

- *Scenario B*: REC load includes the same end-users of Case A, with the addition of member with ID #12.

As shown in Table 1, the electricity consumption of the City Hall (9048 kWh/year) is significantly higher than the average demand of other members (2640 kWh/year). This test aims to assess whether the chosen benefits-sharing model, through appropriate parameter tuning, can adapt to different REC configurations and user compositions. For instance, especially in the case of economic-sharing model purely proportional to the member's energy consumption, the inclusion of a highly energy-intensive user, such as the City Hall in scenario B, may lead to an asymmetry in the revenue distribution, undermining the stability of the community.

Figure 1 shows two violin plots illustrating the monthly energy demand distribution of the REC members. The top graph refers to the configuration without member #12 (scenario A), while the bottom graph includes member #12 in the REC (scenario B). Each colored dot of the violin plot corresponds to the monthly demand of a single user, while the black dot with a white outline indicates the median value of the distribution for that month. In scenario A, all end-users exhibit comparable levels of electricity consumption. Across all the months, members #3 and #8 record the lowest energy demand values (the mean of the lowest monthly energy consumption data is 107 kWh/month), while members #9, #5, and #11 present the highest demand. However, the spread of data, for each month, is low. In scenario B, the inclusion of City Hall (member #12) alters the energy data distribution. Member #12 consistently exhibits the highest energy demand, markedly exceeding the values observed for all other members.

To compute the shared energy within REC, the hourly load curves of end-users are needed. As members have made available only monthly electricity bills, standardized hourly profiles were used to reconstruct the hourly load curves. These profiles are based on percentage coefficients representing the weight of each hour compared to all hours of a day.

Indeed, for all members, except for members #1 and #12, the standardized profiles published by ARERA were adopted, as they offer a higher level of differentiation (e.g., by user type, POD nominal power, etc.) and are available specifically for the province of Benevento, where the REC is located [21]. For members #1 and #12, the load profiles made available by GSE are used, since ARERA profiles do not account for the prosumers equipped with PV systems and do not provide hourly profiles for users with *other usage* electricity contracts.

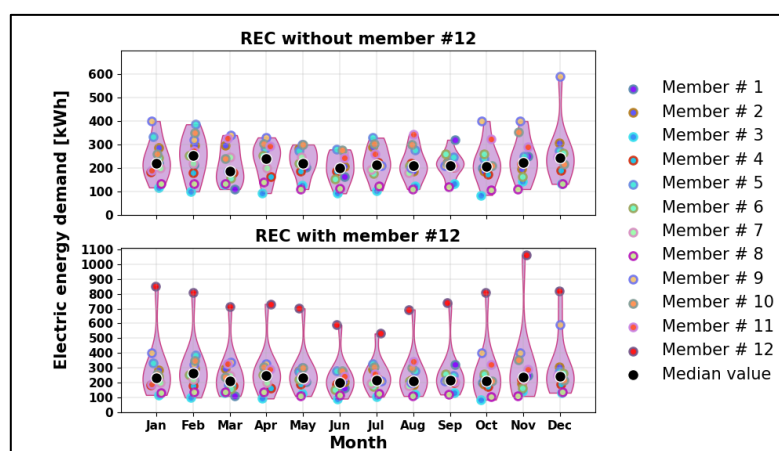


Figure 1. Monthly energy demand distribution of REC members, represented through violin plots. Top: configuration without member #12. Bottom: configuration with member #12.

The REC project involves the installation of PV systems on City Hall and the Municipal sports hall rooftops. Therefore, through PVGIS (Photovoltaic Geographical Information System) [22], the PV production of those plants is simulated. The simulation input parameters are reported in Table 2.

Subsequently, the energy shared among REC members was calculated as shown in (Eq. 2.1):

$$E_{SH}(t) = \min (E_{el,load,REC}(t); E_{el,PV}(t)) \quad (2.1)$$

Where:

- $E_{SH}(t)$: hourly shared energy within REC at hour t;
- $E_{el,load,REC}(t)$: is the REC load at hour t, computed as the sum of energy demand of all REC members (according to the definitions of scenario A and B);
- $E_{el,PV}(t)$: hourly PV production.

Moreover, the energy shared by each REC member is determined considering the balance between PV production and the total REC load, as shown in Eqs. 2.2 and 2.3. When PV production exceeds REC load at a given hour ($E_{el,load,REC}(t)$), the energy shared by each member is set equal to their energy demand.

$$E_{SH,member \#}(t) = E_{el,load,member \#}(t) \quad (2.2)$$

Conversely, if PV production is lower than the total REC load, the energy shared is distributed proportionally among members, based on each member's share of the total demand:

$$E_{SH,member \#}(t) = \frac{E_{el,load,member \#}(t)}{E_{el,load,REC}(t)} \cdot E_{SH}(t) \quad (2.3)$$

Where:

- $E_{el,load,member \#}(t)$: energy demand of member # at hour t;
- $E_{el,load,REC}(t)$: total REC load at hour t;
- $E_{SH}(t)$: total energy shared as defined in Eq 2.1.

2.3. Benefits-sharing model

This section illustrates the mathematical formulation of the benefits-sharing model applied to a real REC in Southern Italy. The first step involves computing the total incentive gained by the REC due to shared energy within the community. According to current regulatory support schemes (Section 2.1), with an hourly time step resolution, the total REC revenue is computed as follows:

Table 2. Orientation and sizing parameters of PV systems installed on each building, including azimuth and tilt angles, and peak power capacity.

Buildings	Azimuth angle [°]	Tilt angle [°]	Peak Power Watts [kWp]
Municipal sports hall	South-West + 64.78°	32	22
City Hall	South-West + 17°	25	7.0

$$G_{REC}(t) = E_{SH}(t) \cdot (TIP(t) + V) \quad (2.4)$$

Where:

- $G_{REC}(t)$: total hourly gain of REC;
- $TIP(t)$: incentive tariff, as defined in Section 2.1;
- V : cost-reflective tariff established by ARERA. For 2022, the reference year for the electricity bills provided, it is set to 0.7780 c€/kWh_{el,SH} [23].

For each member, the share of distributed benefits at each hour is computed as follows:

$$G_{member\#}(t) = \frac{\omega_{member\#}(t)}{\sum_i^N \omega_{member\#i}(t)} \cdot G_{REC}(t) \quad (2.5)$$

Where:

- $G_{member\#}(t)$: incentive allocated to member # at time t;
- $\omega_{member\#}(t)$: weight parameter defining the share of the total REC gain assigned to each user;
- N : number of REC members (11 in scenario A and 12 in scenario B).

The weight parameter ω is the core of the proposed economic-sharing model and is composed of two terms that balance technical performance and social equity:

$$\omega_{member\#} = \beta_1 \cdot B_{1,member\#}(t) + \beta_2 \cdot B_2 \quad (2.6)$$

Where:

- $B_{1,member\#}(t)$ is the *technical component* of model, defined as:

$$B_{1,member\#}(t) = \alpha_1 \cdot E_{el,SH,member\#}(t) + \alpha_2 \cdot E_{el,load,member\#}(t) \quad (2.7)$$

This term considers the member's shared energy and total energy demand. There are two reasons behind this decision: on one hand, members whose loads are aligned with PV generation (and thus who contribute more to shared energy) should be rewarded more, as they help to maximize the REC's incentive revenue. On the other hand, it is crucial to avoid penalizing users with inflexible loads or higher consumption, such as energy-intensive users.

- B_2 is the *solidarity component*, introduced to incorporate social criteria, such as EP. As REC members made the ISEE data unavailable, a proxy indicator was derived from each member's electricity bill, specifically the unit electricity cost, including taxes and VAT (value-added tax), as reported in Table 1. Then, $B_2 \in \{1, 3, 5\}$ factor is associated with each member based on Cu dataset's quartiles: 1 for end-users whose Cu is below the first quartile, 3 for members whose Cu is between the first and second quartile, and 5 for members with Cu above the second quartile. This categorization allows greater redistribution toward users potentially facing energy poverty.
- $\beta_1, \beta_2, \alpha_1$ and α_2 are the tunable coefficients used as degrees of freedom in the model. Their adjustment allows the model to be tailored to the specific structure and objectives of the REC. In this study, the coefficients range as follows: $\alpha_1, \alpha_2 \in [1, 30]$ and $\beta_1, \beta_2 \in [1, 10]$, with a step size of 5 and 1, respectively.

2.4. Equity and Performance Assessment Metrics

In this section, the metrics used to compare the outcomes of different model configurations and the tuning of free parameters ($\beta_1, \beta_2, \alpha_1, \alpha_2$) are described.

The Gini index is widely used in statistical and economic areas to evaluate the asymmetry of resource allocation among a given population [24]. In this case, the resource is the annual income of REC distributed to each member. The Gini index ranges from 0, where every member receives the same amount of revenue (perfect equality), to 1, the worst case (maximum inequality). It is calculated as follows:

$$Gini_{index} = \frac{2 \sum_{i=1}^N i \cdot G_{member\#i}}{n \sum_{i=1}^N i} - \frac{N+1}{N} \quad (2.8)$$

Where $G_{member\#i}$ is the total annual gain received by member #i; N is the total number of members. The Gini index is introduced to assess the fairness of the distribution of benefits. In fact, while it is appropriate to allocate higher gains to members who contribute more to energy sharing or are in potential EP conditions, excessive disparity in benefits distribution may undermine the long-term stability of REC.

To evaluate the social effectiveness of the methodology applied for potential members in EP conditions, the average income received by users in potential EP conditions is compared to the average income of those not in EP conditions.

Finally, a unitary cost of shared electricity (Cu_{SH}) is computed to assess whether the proposed model appropriately rewards users who contribute mainly to energy sharing. This indicator reflects the economic return (€/kWh) per unit of shared energy per member.

3. Results

In the following section, the results obtained are presented. In scenario A, the total annual economic benefit gained by the REC amounts to 979 €, while in the second case it slightly increases to 989 €. The additional member introduced in scenario B does not provide a significant contribution in terms of incentivized shared energy within the REC. Despite its greater energy demand, the amount of shared energy within REC is low.

For each scenario, an optimization algorithm, based on the construction of the Pareto front and the adoption of the Utopia criterion, was applied. The developed algorithm aims to identify the combination of the decision parameters that simultaneously minimizes the Gini index and maximizes the average economic return for users in a potential EP condition.

3.1. Potential EP condition

According to the methodology described in the above section, the end-users in the potential EP condition are identified for both scenarios. In scenario A, five users, specifically members 1, 4, 5, 6, and 8, have a Cu higher than the dataset's median value of 0.396 €/kWh, thus falling under the potential EP condition. In scenario B, the newly added member is also in a potential EP condition, with a Cu of 0.403 €/kWh exceeding the new median value of 0.399 €/kWh. Figure 2 displays the box plots of Cu values for both scenarios. Red markers indicate members as being in a potential EP condition, whereas green markers represent members not falling within this category.

3.2. Pareto Front

According to the prescribed ranges for $\beta_1, \beta_2, \alpha_1, \alpha_2$, 2500 distinct parameter combinations were evaluated. For each combination, the Gini index and the average income received by users in a potential EP condition were computed. The non-dominated solutions were identified through an algorithm developed in the MATLAB environment: 218 and 318 in scenarios A and B, respectively.

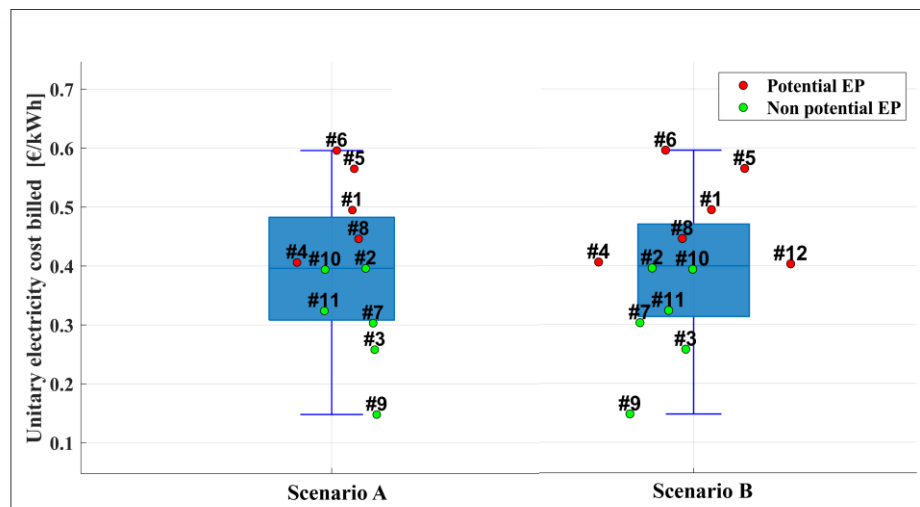


Figure 2. IDs. with Potential EP e IDs with non potential EP.

Figure 3 illustrates the resulting Pareto fronts. The Utopia point corresponds to the hypothetical solution in which each objective function attains its individual optimum. The maximum average income for members in EP condition is 79.1 € in scenario A and 78.8 € in scenario B. Conversely, the minimum Gini index values are 0.131 and 0.132 for scenarios A and B, respectively. It is worth noting that across the non-dominated solutions of both scenarios, the values of singularly optimized objective functions are similar.

For both scenarios, a trade-off between the two objective functions was determined by selecting the solution with the shortest Euclidean distance to the Utopia point. In scenario A, the optimal compromise yields an average EP income of 74.3 € with a Gini index of 0.183. In scenario B, the corresponding values are 71.5 € and 0.199. The higher inequality observed in scenario B is attributable to the inclusion of member #12, whose markedly greater energy demand skews the distribution of benefits. Nevertheless, appropriate tuning of the decision variables mitigates this effect by minimizing the Gini index.

3.3. Average income

To assess the effectiveness of the proposed benefit-sharing model, the annual average income of members in the potential EP and non-potential EP conditions is computed. In Figure 4, the blue bars represent the yearly EP income, while the orange bars refer to non-EP members.

In both scenarios, the optimal solution requires a higher value of α_2 , the decision variable proportional to energy demand. Therefore, since users in an EP condition exhibit lower energy consumption, the sharing model tends to allocate greater benefits to non-EP members (101.2 € in scenario A and 91.7 € in scenario B) than EP members (74.3 € and 73.2 €, respectively). However, by assigning the coefficient of the solidarity term (β_2) 9 times higher than that of the technical term (β_1), the model contains the disparities of the distribution gains. In scenarios A and B, the income gap between non-EP and EP members is limited to 26.9 € and 18.5 €. The reduced gap in scenario B is attributable to the inclusion of a new member who is both in a potential EP condition and characterized by high energy demand, thus benefiting more substantially from the sharing mechanism.

3.4. Unitary cost of shared electricity

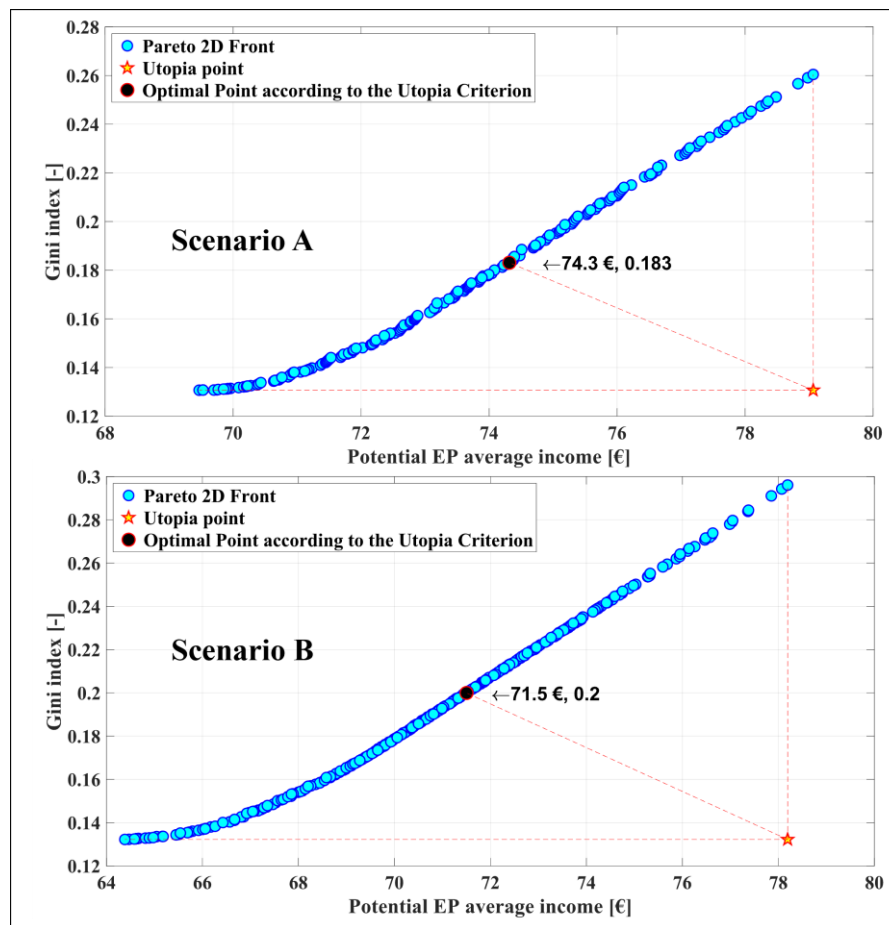


Figure 3. Pareto 2D front of non-dominated solutions, utopia, and optimal point for both scenarios.

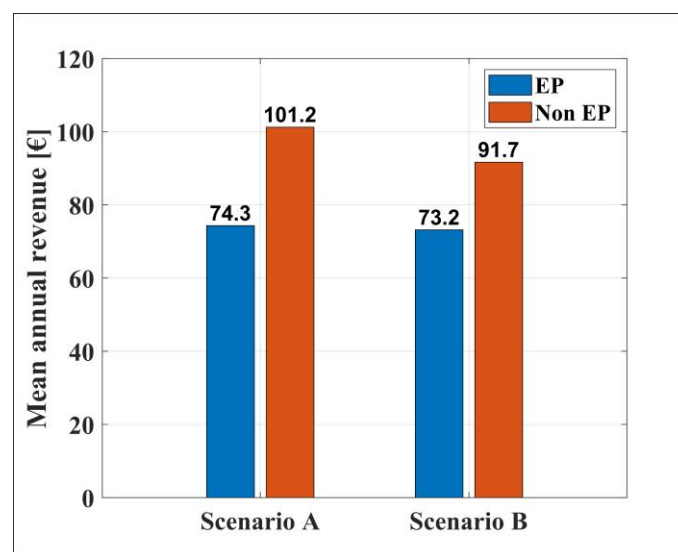


Figure 4. The annual average income of REC members in potential EP and non-EP conditions, under the optimal economic benefit-sharing configuration.

Finally, to assess whether the proposed model adequately rewards users who contribute mostly

Table 3. Summary of the results obtained from the optimization assessment.

Decision variables	Annual average EP income [€]	Gini index [-]
Maximum annual average EP income		
Scenario A		
$\alpha_1 = 1, \alpha_2 = 1, \beta_1 = 1, \beta_2 = 9$	79.1	0.260
Scenario B		
$\alpha_1 = 1, \alpha_2 = 1, \beta_1 = 1, \beta_2 = 9$	78.2	0.296
Minimum Gini Index		
Scenario A		
$\alpha_1 = 1, \alpha_2 = 31, \beta_1 = 3, \beta_2 = 5$	69.5	0.131
Scenario B		
$\alpha_1 = 1, \alpha_2 = 41, \beta_1 = 5, \beta_2 = 9$	64.4	0.132
Optimal solution		
Scenario A		
$\alpha_1 = 1, \alpha_2 = 46, \beta_1 = 1, \beta_2 = 9$	74.3	0.183
Scenario B		
$\alpha_1 = 31, \alpha_2 = 21, \beta_1 = 1, \beta_2 = 9$	71.5	0.199

to energy sharing, the Cu_{SH} is computed. Figure 5 shows the results obtained. The magenta bars represent the annual amount of shared energy per member, while the light blue bars represent the corresponding Cu_{SH} value. In both scenarios, due to lower amounts of shared energy, members in a potential EP condition exhibit higher Cu_{SH} values compared to non-EP members. For instance, members #1 and #8, both identified as EP users, are characterized by a lower amount of shared energy, consequently, their Cu_{SH} is higher compared to Cu_{SH} of other members, especially those who are not in EP condition. A noteworthy exception is member #3: despite sharing a small amount of energy, the corresponding Cu_{SH} (0.06 and 0.07 €/kWh in Scenario A and B, respectively) remains low. This outcome is due to the user falling in the category of users whose Cu value is lower than the threshold of the first quartile, thereby excluding him from the EP category. This result reinforces the notion that the benefit-sharing model is designed with a solidarity-oriented logic, allocating resources preferentially to those with higher vulnerability.

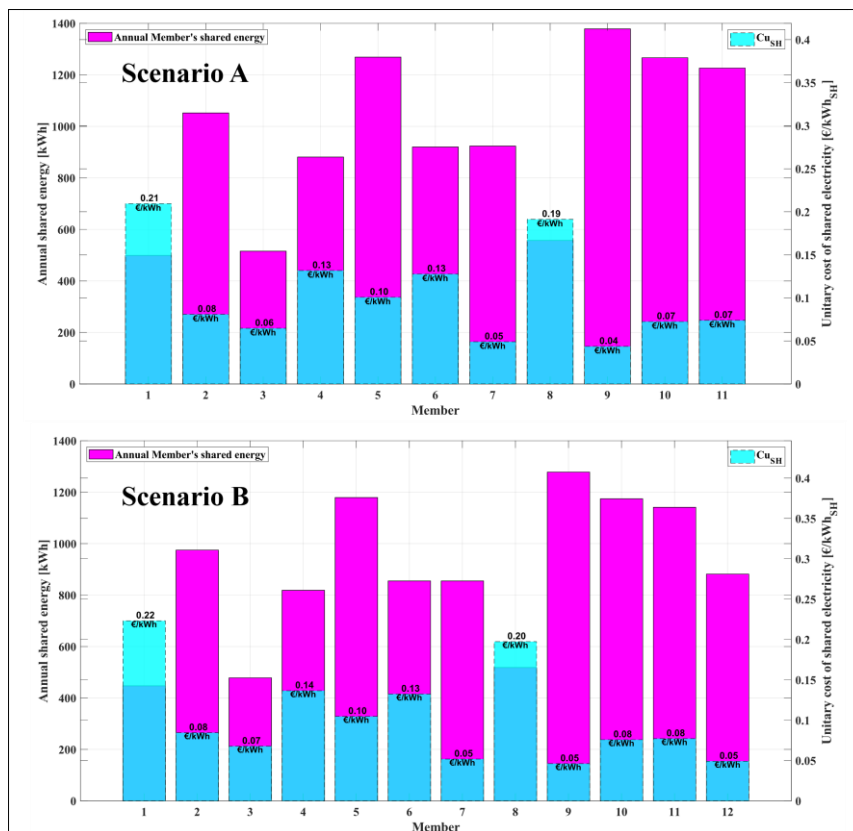


Figure 5. Comparison between the annual shared energy and the unit revenue from shared energy across REC members.

4. Conclusions

This study analyzed the outcomes of applying a novel economic model for redistributing incentives within a REC using a real case study set in Campolattaro. The main objective was to develop a simple yet effective benefit-sharing scheme that was accessible even to non-expert users familiar only with basic spreadsheet tools and applicable in contexts where game-theoretical approaches are unsuitable. The proposed model was also intended to be flexible and adaptable to variations in the composition of the REC, acknowledging that membership is open and subject to change over time.

Two scenarios were analyzed to test the model's adaptability. The first represents the baseline configuration, while the second introduces a member with significantly higher energy demand than the other participants' average. This allowed for the evaluation of the robustness of the benefit-sharing model in the presence of users whose characteristics could distort a purely proportional distribution of incentives.

The model is structured around two main reward logics: one based on energy demand and the other on the socioeconomic vulnerability of the users, specifically targeting those in potential energy poverty conditions. A range of values was defined for the four decision variables that weigh these two components, and approximately 2500 combinations were examined for each scenario. For every combination, two key outcome indicators were calculated: the Gini index, to

measure inequality in benefit distribution, and the average income received by users in the EP condition. These two functions were chosen to assess the model's performance and guide the search for optimal configurations. Through this analysis, the Pareto fronts for both scenarios were constructed, and optimal solutions were identified based on their proximity to the so-called Utopia Point. The comparison between scenarios showed that, although the presence of a high-consumption user introduces a tendency toward greater inequality, the tuning of the decision parameters allows the model to preserve fairness. Specifically, by increasing the solidarity coefficient β_2 to a value nine times greater than the technical one (β_1), the model successfully limited the income gap between EP and non-EP members to about €27 in Scenario A and €19 in Scenario B.

To further assess the model's equity orientation, the annual income of users was broken down by EP status, and their unitary remuneration per kilowatt-hour of shared energy was analysed. The results show that, despite non-EP users receiving higher absolute annual revenues, mainly due to their greater energy consumption, the users in EP conditions benefited from a higher revenue per unit of shared energy. This demonstrates the model's capacity to reward vulnerable users without undermining the incentive mechanisms linked to individual contributions.

In the current version of the model, potential EP conditions were assessed based on the unit cost of electricity in users' bills. A more refined classification could be achieved using ISEE or other composite indicators for future developments. Additionally, the current attribution of scores to users in the EP condition is based on a step function using quartile thresholds, resulting in three discrete classes. Future iterations of the model could implement a continuous scoring function or increase the number of classes (e.g., from three to four or five) to improve the accuracy and fairness of the redistribution.

In conclusion, the proposed model provides a practical, adaptable, and socially mindful approach to sharing economic benefits in energy communities. It proves effective in reconciling equity with efficiency, and its simplicity makes it suitable for real-world application, even in self-managed REC settings. The results confirm that, with appropriate tuning of its parameters, the model can support inclusive energy transitions, ensuring that the most vulnerable members are not left behind.

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